Goal: decide whether, when, and how to fuse frozen DNA, RNA, and protein encoders for a biological task, and leave a method that predicts it before training.
Fusion has to beat strong single-encoder and simple-fusion baselines; different representations are not by themselves independent information.

Stage 1 Representation analysis, done

What does each frozen encoder carry, and how do their spaces relate?

12 encoders, 2 datasets, 3 tracks.
1. Protein and codon-level encoders carry the stability signal; nucleotide encoders carry GC3, which translation erases.
2. Most raw agreement is composition; only within-family and codon-with-protein pairs stay aligned after the control.
3. Alignment never predicts error; gains appear where alignment is low.
4. Four length features match all twelve encoders on GTEx.

Redundant families found: ESM-2; RNA-FM with RiNALMo.

Forces: a biology-specific definition of alignment; a baseline hierarchy; a pruned set of pairs to fuse first.

Stage 2 Define alignment (decided)

What should alignment mean here, and should the fusion models enforce it?

Alignment = agreement on what the central dogma shares, after composition is removed. Positional alignment (the codon grid) only on coding sequence; a token-location step. No alignment loss by default. Alignment is measured and reported beside every result; fusion benefit is a separate held-out gain, Δfusion.
Literature: alignment theory (redundancy vs uniqueness); MSAlign (alignment as retrieval); sequence-structure alignment (SA-PLM, OmniGenome) is a different sense of the word.

Forces: architectures that let modalities interact without forcing similarity.

Stage 3 Fusion design

How do we combine the encoders, given the Stage 2 decision?

The research plan's baselines first: item 1, BioLangFusion reproduced exactly on identical splits, then its encoders swapped; item 2, LucaOne (joint pretraining, not fusion); IsoFormer on GTEx. Floor: concatenation + MLP. Then the project's designs: Coupled Mamba on the codon grid (coding sequence; interaction through the recurrence, no forced similarity), then IsoFormer-style cross-attention (GTEx; no grid needed).

Forces: every design is scored against the plan's baselines on identical splits.

Stage 4 Experiments, benchmarking

Does fusion help, for which task, and could Stage 1 have told us in advance?

The research plan first, in its order: (1) BioLangFusion reproduced, then encoder swaps; (2) LucaOne; (3) half-life; (4) translation efficiency; (5) mRFP control. Then Coupled Mamba and cross-attention against all of those baselines; ablations (alignment loss, grid); the method test: do Stage 1 metrics predict Δfusion? Last: (6) structure, optional; (7) distillation, exploratory. COMET adds multi-molecule tasks; mRNABench adds homology-aware splits.

Delivers: a tested decision method, or evidence that frozen geometry does not transfer.

Evaluation rules from Fusion or Confusion on every comparison: identical heads, tuned baselines, grouped cross-validation, mean ± SD over seeds, a statistical comparison.